

Exploring FIA Data: Two approaches to modeling

Tatum Beaugh | Ashna Coondiah | Erin Fennema | Keegan Fontaine |
Anvi Kothari | Pawel Nowak | Liza Shchekhlik

1. Introduction

What is FIA?

The USDA Forest Service Forest Inventory and Analysis (FIA) Program is a multi-decade national initiative that monitors the health, composition, and change of forests across the United States, essentially serving as the forest equivalent of the U.S. Census Bureau.



Scan to visit FIA site

How is FIA Organized?

FIA data are collected at permanent sample plot locations, producing one of the largest and most comprehensive forest datasets in the country.

- Each plot contains many trees
- Data is recorded on both a *tree* and *plot* level

How to Handle FIA's Scale?

Given the many different ways of exploring FIA data, our project explores two complementary analytical approaches:

- A machine-learning library to estimate FIA variables **nationwide**
- A specific model and paired dashboard to estimate **forest mortality values** across the U.S.

2.1. Methodology

Modeling Library Team

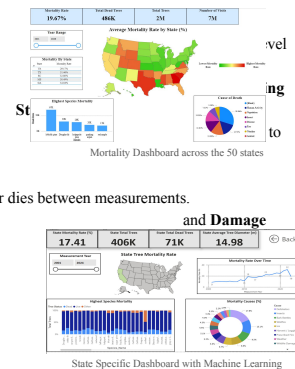
- **Motivation:** Our team aimed to find a way to accurately assess forest attributes like biomass and growth. FIA plots are spread out farther than the local precision that is needed for small-area planning and reporting.
- We developed a Streamlit and Leaflet-based application, enabling users to use the official FIA **dataset** of forest/non-forest plots.
 - Forestry label is **determined by total proportion of forest land condition exceeding a threshold of 10%**
- The app trains a grid of machine learning (ML) models across the Continental USA, utilizing **satellite data** as explanatory characteristics.
 - Satellite data is sourced from **Google Earth Engine's Satellite Embeddings dataset**.
 - Embeddings importance analysis proved we don't need all satellite data to get similar results - out of 64 we only need a fraction of them.
- Users can then interactively **define a local area to model any target variable**.
 - The app gives the users the flexibility to configure the training process as needed,
 - The closest model (i.e. the closest to the area) is found, trained with "nearest" balanced forest/non-forest records, and predictions are generated into exportable forms (graphs, tables, etc.).
 - The user might want to model a FIA metric (e.g. Carbon) or any GEE characteristic from Google's catalogue.



2.2. Methodology

Forest Mortality Team

- **Motivation:** Tree mortality is a critical indicator of forest health, ecosystem resilience, and carbon stability. Understanding the patterns of mortality helps inform sustainable forest management and climate adaptation strategies.
- **Data Source:** FIA data are collected at permanent sample plots across the United States and include detailed measurements on tree characteristics, species identity, mortality status, and recorded causes of death.
- **Plot-Level Modeling & Dashboard Framework**
 - To examine forest trends, we aggregated individual-tree outcomes to mortality rates (0–1) across measurement cycles.
 - Preprocessing steps included **Filtering** remeasurement plots, **values**, Encoding **categorical variables** and
 - Final model outputs were integrated into an interactive Power BI dashboard, explore mortality patterns by year, state, forest type, and ownership.
- **Individual-Tree Mortality Modeling**
 - We trained a logistic regression model to predict whether an individual tree survives or dies between measurements.
 - Predictors included **Diameter and growth, Crown condition, Species identity indicators**.
 - Model coefficients were reported as **odds ratios**, allowing us to interpret how increases or decreases mortality risk.
 - We additionally fit:
 - **Cause-specific models** (insects, disease, weather, competition, harvesting)
 - **Species-specific models** to identify mortality drivers within individual species



3.1 Practical Outcomes

Modeling Library Team

The FIA provides long-term data on forest conditions across the U.S., but its plot-based design limits observations to specific locations, leaving gaps between sampled plots. To overcome this, we extended FIA plot data with satellite imagery from Google Earth Engine, enabling continuous nationwide estimates of forest variables. We built a scalable machine learning library that generalizes beyond individual plots.

3.2 Practical Outcomes

Forest Mortality Team

We built a pipeline processing FIA data across all 50 states to predict whether individual trees survive between measurement cycles. Outputs were integrated into an interactive Power BI dashboard where users can explore mortality patterns by year, state, forest type, and ownership. We also fit models targeting specific causes of death — such as insects, disease, and weather — and individual species, helping pinpoint what is driving tree loss in different regions across the U.S.

4. Next Steps

Future work will expand our modeling framework by comparing logistic regression with more flexible machine learning approaches and incorporating climate and disturbance variables to better capture environmental drivers of mortality. We also plan to develop spatial mortality risk maps and extend the models to forecast future mortality under changing climate conditions. Finally, we aim to enhance the dashboard as a decision-support tool for forest managers and policymakers.

5. Acknowledgements

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